2.6 Encryption

You have learnt a lot about how data is transmitted in this chapter. You have also learnt why it is important for data to be received correctly and how errors can be detected to make sure this happens. You also need to understand that the data that is transmitted often needs to be protected during transmission. This is because data is valuable, especially when it contains our personal details and information that is personal to us. A hacker may try and steal the data during transmission so that they can use it for their own gain. One method of protection that can be used when transmitting data is **encryption**.

Data before encryption takes place it is called **plain text**. An encryption algorithm, called an **encryption key**, is used to scramble the data and make it meaningless. This meaningless data is called **cipher text**. The cipher text can then be transmitted from one device to another. The receiving device uses the key to decrypt the cipher text and return it to its plain text form. This means the data will have meaning again if read.

One important thing to note about encryption is that is does not stop a hacker stealing the data that is transmitted. It just means that the data that is stolen will be meaningless, as it will be encrypted and therefore scrambled. They will not understand any of the data in the file.

There are two main methods of encryption that can be used **symmetric** and **asymmetric**.

The process for symmetric encryption is:

- Plain text is encrypted into cipher text using an encryption key.
- The cipher text and the encryption key are sent separately to the receiving device.
- The same key is then used to decrypt the cipher text back into its plain text form.

This is the simplest method of encryption and is still used for data that is not of a high level of importance. It is sometimes used by people just wanting to encrypt the data that they store on their own computer or external storage device, as it isn't being transmitted across a network. However, people began to worry that it could be too easy for a hacker to intercept both the cipher data and the encryption key if it is sent across a network, or the internet, so a more secure method was developed called asymmetric encryption.

The process for asymmetric encryption is:

- Plain text is encrypted into cipher text using a public key. This is also a type of encryption algorithm.
- The cipher text is transmitted to the receiving device.
- The cipher text cannot be decrypted using the public key, it is decrypted using a private key.

A person who uses an asymmetric method of encryption has a public encryption key. This is called a public key because it is made public for the encryption process. Making something public means that anyone can see and use it. Any device that wants to send you encrypted data may obtain your public key and encrypt the data using your public key. You also have a private key. Any encrypted data that your device receives that has been encrypted with your public key can be decrypted with your private key. They act as a pair. Your private key is always kept private though, your device does not make it available to any other unless authorisation is given to do so.

KEY WORDS

encryption: a method of securing data for storage or transmission that scrambles it and makes it meaningless.

plain text: the name given to data before encryption.

encryption key: a type of algorithm that is used to encrypt data.

cipher text: the name given to data after transmission.

symmetric: a type of encryption that uses the same key to encrypt and decrypt data.

asymmetric: a type of encryption that uses two different keys to encrypt and decrypt data.

The process can work in reverse as well (a private key is used to decrypt and a public key is used to decrypt), and this is what can occur in processes such as digital signatures. These are used in security protocols such as SSL (secure sockets layer). You will learn more about these in Chapter 5, about applications of the internet.

COMPUTER SCIENCE IN CONTEXT

A bank uses asymmetric encryption for all the banking transactions that are completed on a daily basis. As you can imagine, the data that is transmitted for these banking transactions is very sensitive and personal data, so the bank has to make sure that they use a very secure method of encryption for this process.

Questions

24 What is the name given to an encryption algorithm?

25 What is the name given to data that has been encrypted?

26 Does encryption stop data being stolen by a hacker? Why or why not?

27 Why is asymmetric encryption a more secure method of encryption than symmetric?

SUMMARY

Data is broken down into packets to be transmitted from one device to another.
Each packet contains three parts: the packet header, the payload and the trailer.
The packet header includes the destination address, the packet number and the originator's address.
The payload is the data the sender wants to transmit.
The trailer contains data such as the error detection method to be used.
A process called packet switching can be used to send the data from one device to another across a network.
Data is transmitted using serial or parallel transmission.
Data is also transmitted using simplex, half-duplex or full-duplex transmission.
An interface called USB can be used to transmit data. This is often used to connect hardware such as a keyboard to a computer.
Errors can occur when transmitting data due to interference.
Methods are required to detect any errors in transmission.
A parity check is an error detection method that uses a parity bit to detect errors. An odd or even parity check method can be used.
A checksum is an error detection method that uses a calculated value to detect errors.
An echo check compares data that is sent and received to see if they match, to detect errors.